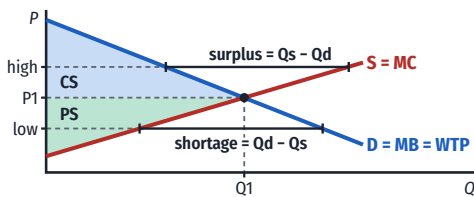
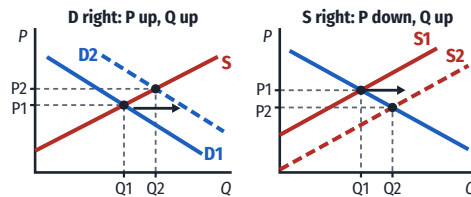


1 Market equilibrium and surplus

Label: Write $D = MB = WTP$, $S = MC$, P_1 , Q_1 , CS and PS . At a given price, Q_d is what buyers want and Q_s is what sellers offer.

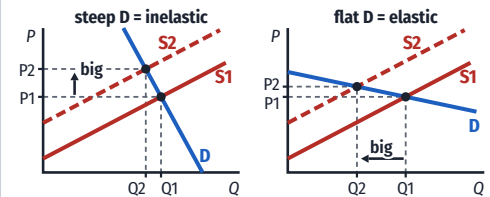
1. Draw D and S crossing at P_1 , Q_1 . Shade CS and PS .

- $CS = \frac{1}{2} \times Q_1 \times (\text{the price where } D \text{ hits the } P \text{ axis} - P_1)$.
- $PS = \frac{1}{2} \times Q_1 \times (P_1 - \text{the price where } S \text{ hits the } P \text{ axis})$.
- At Q_1 , $CS + PS$ is the largest it can be, so DWL is 0.
- Above P_1 , sellers offer more than buyers want, so the price falls.
- Below P_1 , buyers want more, so the price rises.

2 Shifting demand and supply

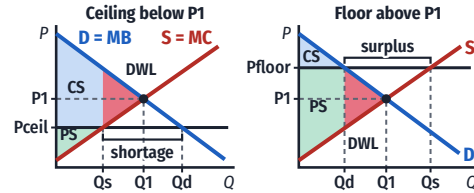
Label: Mark D_1 , D_2 (or S_1 , S_2), the arrow, and P_1 , Q_1 , P_2 , Q_2 .

- Buyers' income or tastes shift D ; sellers' costs shift S .
 - Draw the shift with an arrow and mark the new P_2 and Q_2 .
- If D shifts right, P and Q both rise; left does the opposite.
 - If S shifts right, P falls and Q rises; left does the opposite.
 - If both shift, only P or only Q is certain. For D right and S left, P rises; the change in Q depends on which shift is bigger.
 - A change in the good's own price is a move along the curves.

3 Elasticity: steep versus flat demand

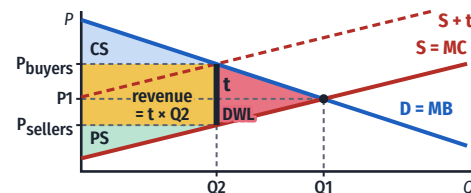
Label: Mark steep D , flat D , S_1 , S_2 , P_1 , Q_1 , P_2 and Q_2 .

- Draw the same shift from S_1 to S_2 on both graphs.
 - Steep D makes P change a lot and Q a little; flat D does the reverse.
- Elasticity $\epsilon = \% \text{ change in } Q \div \% \text{ change in } P$. Use its size $|\epsilon|$.
 - On a straight D , ϵ at a point $= (1 \div \text{slope of } D) \times (P \div Q)$.
 - If $|\epsilon| > 1$, demand is elastic. If $|\epsilon| < 1$, demand is inelastic.
 - Raising P raises spending $P \times Q$ if demand is inelastic.

4 Price ceiling and price floor

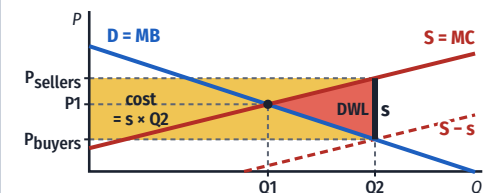
Label: P_{ceiling} is a legal top price; P_{floor} is a legal bottom price.

- Draw D and S (neither curve shifts), then a flat line at the legal price. Read Q_s on S and Q_d on D at that price.
- Only the smaller amount is sold: Q_s with a ceiling (shortage $= Q_d - Q_s$), and Q_d with a floor (surplus $= Q_s - Q_d$).
 - $DWL = \frac{1}{2} \times (Q_1 - \text{the amount sold}) \times (\text{the gap between } D \text{ and } S \text{ at the amount sold})$.
 - With a ceiling, sellers lose. Buyers gain on price but lose on the units no longer sold.

5 Per-unit tax

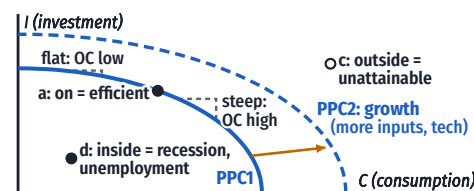
Label: Mark $D = MB$, $S = MC$, $S + t$, P_1 , Q_1 , Q_2 , P_{buyers} and P_{sellers} .

- Draw D and S crossing at P_1 , Q_1 , then $S + t$ (moved up by t).
 - $S + t$ meets D at Q_2 . Read P_{buyers} on D and P_{sellers} on S at Q_2 .
- Tax revenue $= t \times Q_2$, and $DWL = \frac{1}{2} \times t \times (Q_1 - Q_2)$.
 - Whoever has the steeper curve pays more of the tax.
 - Buyers pay $P_{\text{buyers}} - P_1$ more; sellers get $P_1 - P_{\text{sellers}}$ less.
 - If buyers legally pay the tax, shift D down by t instead. The result is the same.

6 Subsidy

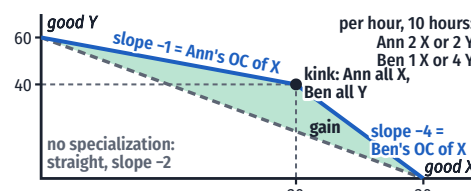
Label: Mark $D = MB$, $S = MC$, $S - s$, P_1 , Q_1 , Q_2 , P_{buyers} and P_{sellers} .

- Draw D and S at P_1 , Q_1 , then $S - s$ (moved down by s).
 - $S - s$ meets D at Q_2 . Read P_{buyers} on D ; sellers get $P_{\text{buyers}} + s$.
 - Shade the cost box (0 to Q_2) and the DWL from Q_1 to Q_2 .
- Government cost $= s \times Q_2$, and $DWL = \frac{1}{2} \times s \times (Q_2 - Q_1)$.
 - Units past Q_1 cost more than buyers value them, causing DWL .
 - Buyers' and sellers' gains = government cost - DWL .

7 PPC (production possibilities curve)

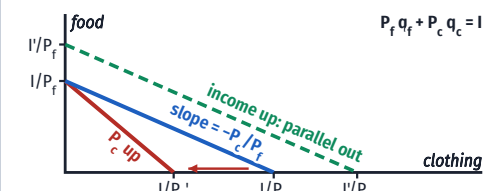
Label: Put C (consumption) and I (investment) on the axes.

- Draw a bowed-out PPC ; mark a on it, d inside, and c outside.
 - More workers, machines, or tech push it out to PPC_2 .
- The OC (opportunity cost) of one more C is the slope's size. The OC rises as you make more C , so the PPC bows out.
 - A point inside means unemployment or recession, not an inward shift. More I today pushes the PPC out later.

8 Comparative advantage: joint PPC

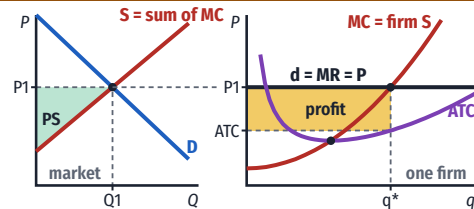
Label: Mark the intercepts, the kink, both slopes, and the gain.

- Each intercept is everyone's total: 60 Y or 30 X in 10 hours.
 - Start at 60 Y . Ann (lower OC of X , slope -1) makes X first, up to the kink; then Ben (slope -4) takes it to 30 X .
- OC of $X = Y$ per hour $\div X$ per hour, so Ann's is 1 and Ben's is 4.
 - Lower OC of X means comparative advantage in X .
 - With more people, add pieces from lowest OC of X to highest.

9 Budget line and spending rule

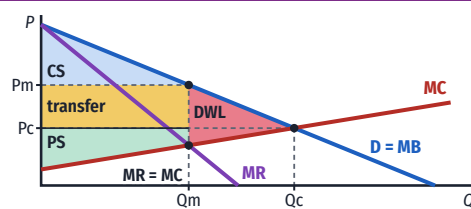
Label: I is income, P_f and P_c are prices, and q_f and q_c are amounts bought.

- Join $I \div P_f$ on the food axis to $I \div P_c$ on the clothing axis.
 - More income (I') shifts it out in parallel; higher P_c pivots it in.
- Rational Spending Rule: buy until $MU_f \div P_f = MU_c \div P_c$, where MU is the extra satisfaction from one more unit.
 - If P_c rises, $MU_c \div P_c$ falls, so buy less clothing.

10 Competitive firm and market

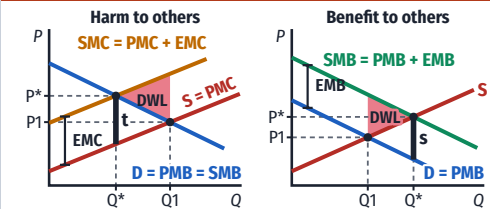
Label: Mark D , S , P_1 , Q_1 and the firm's $d = MR = P$, MC , ATC and q^* .

- Carry P_1 from the market across as the firm's flat line $d = MR = P$.
 - The firm makes q^* where $MC = P$. MC cuts ATC at its lowest point.
- The firm is too small to move P , so $MR = P$ and its MC curve is its supply curve. Market S adds up all firms' MC .
 - Profit $= (P - ATC) \times q$. In the long run, entry pushes P down to the lowest ATC , so profit $= 0$.
 - Profit also equals PS - fixed cost.

11 Monopoly

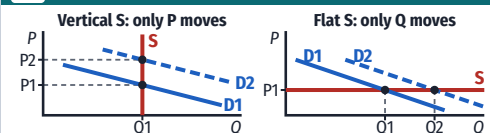
Label: Mark $D = MB$, MR , MC , Q_m , P_m , Q_c , and the DWL .

- Draw D , MR (twice as steep, same P intercept), and MC .
 - Find Q_m where $MR = MC$, then go straight up to D to read P_m .
 - D meets MC at Q_c and P_c . Shade the DWL from Q_m to Q_c .
- If D is $P = a - bQ$ (a = intercept, b = slope), then $MR = a - 2bQ$.
 - The transfer is $P_c - P_m$ as P_c and P_m that becomes profit.
 - Profit $= \text{transfer} + PS = \text{revenue} - \text{the area under } MC$.
 - If MC is flat, $a = 2 \times P_m - MC$ and $b = (P_m - MC) \div Q_m$.

12 Externalities: tax or subsidy

Label: Write $SMC = PMC + EMC$ and $SMB = PMB + EMB$.

- The market makes Q_1 , where D meets S . The best amount Q^* is where the social curve crosses the other curve.
 - Shade the DWL between Q_1 and Q^* ; its point is at Q^* .
- Harm: Q_1 too big. A Pigouvian tax $t = EMC$ at Q^* fixes it.
 - Benefit: Q_1 too small. A Pigouvian subsidy $s = EMB$ at Q^* fixes it.
 - If EMC or EMB is a constant amount, $DWL = \frac{1}{2} \times \text{that amount} \times (\text{the gap between } Q_1 \text{ and } Q^*)$.

13 Extreme slopes: vertical or flat

- If one curve is vertical, that side pays all of a tax and $DWL = 0$. If one curve is flat, the other side pays all of it.

Key: P price · Q quantity · q one firm's quantity · 1, 2 before and after a change (P_1 , Q_1 , D_1 , S_1 , PPC_1) · D demand · S supply · MB marginal benefit (value of one more unit) · WTP willingness to pay · MC marginal cost (cost of one more unit) · MR marginal revenue (extra revenue from one more unit) · MU marginal utility (extra satisfaction from one more unit) · ATC average total cost (total cost $\div q$) · CS consumer surplus (buyers' value minus price) · PS producer surplus (price minus MC) · DWL deadweight loss (surplus lost) · OC opportunity cost (what you give up) · PPC production possibilities curve · C , I consumption, investment (graph 7) · X , Y any two goods · ϵ elasticity, $|\epsilon|$ its size (no minus sign) · $\% \Delta$ percent change · t tax per unit · s subsidy per unit · Q_s , Q_d (or Q_s , Q_d) quantity supplied, demanded · P_{buyers} , P_{sellers} price buyers pay, sellers keep · P_{ceiling} , P_{floor} legal maximum, minimum price · Q^* , P^* efficient (best for society) quantity, price · Q_m , P_m monopoly quantity, price · Q_c , P_c competitive quantity, price · PMC , SMC , EMC private, social, external marginal cost · PMB , SMB , EMB private, social, external marginal benefit · I income (graph 9) · P_f , P_c , q_f , q_c price and amount of food, clothing · I' , P'_c new income, new price of clothing · MU_a , P_a marginal utility and price of good a · a , b intercept and slope in $D: P = a - bQ$ · q^* one firm's best output · d the demand one small firm faces · SID student ID

Symbols: P price · Q quantity · 1 before, 2 after · D = MB (marginal benefit) = WTP (willingness to pay) · S = MC (marginal cost) · CS, PS consumer, producer surplus · DWL deadweight loss · OC opportunity cost · graph N = side 1 · other symbols: each panel's Terms line

How to answer (his grading)	
If the question asks	Do this
"Explain" or "Draw"	Answer first, then give the reason in 2 to 6 sentences. On a graph, label both axes and every curve.
Why does demand slope down, and supply up?	Each extra unit adds less satisfaction, so MB falls as Q rises . Each extra unit costs more to make, so MC rises .
Why is producer surplus a surplus?	Each unit sells for P but costs only its own MC, so sellers keep P minus MC.

Demand = MB = WTP (willingness to pay). Supply = MC.

Opportunity cost and the PPC		graph 7
• Terms: OC = opportunity cost, the value of the best option given up. Sunk cost = money already spent and gone. PPC = the most of two goods you can make.		

If the question asks	Do this
OC of 1 X? (you can make a X or b Y)	Divide. The OC of 1 X is $b \div a$ units of Y.
What does going to an event cost you?	Add hours × your wage (about \$36 an hour) and the ticket's resale value. The price you paid is sunk, so leave it out.
What does college cost?	Tuition, fees, books, and the pay you give up by not working (about \$60,000 a year) count. Rent and food do not, since you pay them anyway.
Why is the PPC bowed out? Where is a recession?	OC rises because the lowest-OC resources switch first. A recession is a point inside it, with no shift.

Comparative advantage		graph 8
• Terms: absolute advantage = making more per hour. Comparative advantage = having the lower OC.		

If the question asks	Do this
Who should specialize in what, and why?	Divide each person's Y per hour by their X per hour: that is their OC of 1 X. Whoever has the lower OC makes X.
Everyone's output of one good doubles. What changes?	Nothing changes. The OCs keep their order, so nobody switches.
What trade price helps both people?	Any price between the two OCs works.

The lower OC decides, not who makes more per hour.

Supply and demand		graphs 1, 2
• Terms: normal good = bought more as income rises (inferior good: less). Substitute = used instead; complement = used with it.		

If the question asks	Do this
What shifts demand right? (The reverse shifts it left.)	Higher income (normal good), lower income (inferior good), a pricier substitute, a cheaper complement, more buyers, a taste for it, or an expected price rise all shift it right.
What shifts supply right? (The reverse shifts it left.)	Cheaper inputs, better technology, more sellers, a tax or rule cut, or an expected price fall all shift it right.
Was it a supply or a demand shock?	If P and Q moved the same way, demand shifted. If they moved opposite ways, supply shifted.
"The price rose, so demand fell." True?	False. A change in the good's own price is a move along demand, not a shift.
Both curves shift. What happens?	If both shifts push P (or Q) the same way, that change is sure. The other depends on which shift is bigger.

Households and the spending rule		graph 9
• Terms: MU_a = the extra satisfaction from one more unit of good a. P_a = the price of good a.		

If the question asks	Do this
Is she spending rationally? (MU and prices given)	Compute $MU \div P$ for each good. Equal means yes. If not, she buys more of the good with the higher $MU \div P$, less of the other.
P_a rises or falls. What should she do?	If P_a rises, she buys less a and more b. If P_a falls, she buys more a and less b.
A heatwave raises MU of a good?	Its demand shifts right, so she buys more of it and less of the other good.

Rational Spending Rule: $MU_a \div P_a = MU_b \div P_b$, so the last dollar on each good adds the same satisfaction.

• **TRAP** Mistake: buying more of a raises MU_a . Correct: it lowers it.

Elasticity		graph 3
• Terms: $\epsilon = \% \Delta Q \div \% \Delta P$, where $\% \Delta = (\text{new} - \text{old}) \div \text{old}$. Elastic means $ \epsilon > 1$ and inelastic means $ \epsilon < 1$.		

If the question asks	Do this
Elasticity? Elastic or inelastic?	Divide $\% \Delta Q$ by $\% \Delta P$, using the starting values. Write " $\epsilon = -0.5$, so inelastic."
ϵ at a point on a straight line?	Use $\epsilon = (1 \div \text{slope}) \times (P \div Q)$. It is elastic above the midpoint and inelastic below.
Price rises. Does spending rise or fall?	It rises if demand is inelastic and falls if elastic. A price fall does the opposite. For big changes, compare $P \times Q$ before and after.
Given ϵ and $\% \Delta Q$, find $\% \Delta P$.	Use $\% \Delta P = \% \Delta Q \div \epsilon$. If $\epsilon = -0.4$ and Q falls 2%, price rises 5%.
Steep or flat curve?	Demand is steep (inelastic) with no close substitute or in the short run. It is flat with close substitutes or in the long run.

Surplus, efficiency, DWL		graph 1
• Terms: WTP = the most a buyer would pay. CS = $WTP - P$ and PS = $P - MC$, summed over units traded. DWL = surplus lost versus the efficient Q, where $CS + PS$ is largest.		

If the question asks	Do this
Why is this area producer surplus?	Each unit sells for P but costs only its MC. The area under MC is variable cost, so PS is what sellers keep.
Why is DWL a "loss" if nobody pays it?	It is the surplus from units no longer traded (Q2 to Q1), each worth more to buyers than its MC. Nobody gets it; tax revenue is a transfer, not a loss.
Is the outcome efficient?	Yes, if nothing distorts it. $MB = MC$, the highest-WTP buyers buy, and the lowest-MC sellers sell. It can still be unfair.

Price controls		graph 4
• Terms: A ceiling is a legal maximum price. It matters only if it is below P1. A floor is a legal minimum price, like a minimum wage. It matters only above P1. Q_s and Q_d are what sellers offer and buyers want at the legal price.		

If the question asks	Do this
Does either curve shift?	No. Neither curve shifts. Draw a flat line at the legal price (graph 4).
How much is actually sold?	The smaller amount. With a ceiling, sellers offer only Q_s , so Q_s sells and the shortage is $Q_d - Q_s$. With a floor, buyers want only Q_d , so Q_d sells and $Q_s - Q_d$ goes unsold.
CS, PS, DWL under a ceiling?	Only Q_s trades, so measure every area out to Q_s . DWL is the triangle from Q_s to Q1 (numbers: see Number problems).
Do buyers gain from a ceiling?	Maybe. Those who still buy pay less, but fewer units sell. Sellers always lose.
Why is the real loss bigger than the DWL?	The triangle assumes the highest-WTP buyers get the units. With a ceiling, luck or lines decide. With a floor, the lowest-MC sellers may not get to sell.
Gouging ban, or a low price with a purchase limit?	It acts like a ceiling: a shortage results, and the limit decides who buys. The model calls this wasteful. It ignores that price jumps feel unfair, and that a cap stops a seller with market power from overcharging.

Taxes and subsidies		graphs 5, 6
• Terms: t and s are the tax and subsidy per unit, and Q2 is the new quantity. P_{buyers} is what buyers pay, and $P_{\text{sellers}} = P_{\text{buyers}} - t$ is what sellers keep.		

If the question asks	Do this
Find Q2 and both prices.	Draw graph 5. Solve demand $P = \text{supply } P + t$ to get Q2 and P_{buyers} .
Does it matter who pays it?	No. Taxing buyers or sellers gives the same Q2 and prices.
Who bears more of the tax, and why?	The steeper (more inelastic) side, since it can't walk away. Flat supply or vertical demand puts it all on buyers. Vertical supply or flat demand puts it all on sellers.
Revenue and DWL?	Revenue = $t \times Q2$. DWL = $\frac{1}{2} \times t \times (Q1 - Q2)$, zero if a curve is vertical. There is no misallocation: price still decides who buys.
Subsidy?	Draw graph 6. Solve demand $P + s = \text{supply } P$. Cost = $s \times Q2$. DWL = $\frac{1}{2} \times s \times (Q2 - Q1)$: extra units cost more than they are worth.

Competitive firms		graph 10
• Terms: A price taker is too small to change P, so its MR (extra revenue per unit) equals P. q is one firm's quantity. Fixed cost is paid even at zero output. Profit is PS minus fixed cost.		

If the question asks	Do this
How much does a firm make?	It makes the q where $P = MC$. Set P equal to the MC formula and solve for q (graph 10).
Should it enter? What about the long run?	Enter if PS is bigger than fixed cost, since profit is then positive. Firms enter or leave until profit is zero.
A cost rises for every firm. What happens to one firm?	Its MC rises more than P does, so it makes less and its profit falls. Some firms then leave.

Monopoly		graph 11
• Terms: Market power means a firm can raise P without losing every buyer. Qm and Pm are its quantity and price. Qc is the efficient quantity, where D meets MC.		

If the question asks	Do this
Why is MR below P?	It must cut the price on every unit to sell one more, so it gains less than P.
Find its Q and P.	Draw MR starting where D hits the P axis, but twice as steep. Qm is where $MR = MC$. Pm is the height of D above Qm (graph 11).
Why does it stop at Qm?	Selling one more unit means cutting P on every unit, so MR falls below P. Past Qm, MR is below MC even though buyers value those units above MC.
Is there DWL?	Yes. It stops at Qm, short of Qc, to keep P above MC. DWL is the triangle between D and MC from Qm to Qc.
Buyers get a subsidy s per unit.	D and MR shift up by s. At the old Q, profit rises by $s \times Q$ and CS stays the same. But $MR = MC$ now at a bigger Q, so Q rises and DWL shrinks.
Does it have a supply curve?	No. It picks one point on each demand curve.

Externalities		graph 12
• Terms: An externality is a cost or benefit to outsiders that the price ignores. PMC is the seller's own cost (S). EMC is the harm to others per unit. SMC = $PMC + EMC$. EMB and SMB are the same for benefits ($D = PMB$). Q1 is the market quantity, Q* the efficient one. A Pigouvian tax is a per-unit tax equal to EMC at Q*.		

If the question asks	Do this
Compare the market Q and the best Q.	Q1 is where D meets PMC. Q* is where $SMB = SMC$. A harmful externality makes Q1 too big; a helpful one, too small (graph 12).
Is the best amount of pollution zero?	Usually no. The first units are worth more than the harm they do, so Q* (where $SMC = SMB$) is above zero.
How big is the DWL?	It is the triangle between SMC and D from Q* to Q1: units that cost society more than they are worth.
What tax fixes it? (Pigouvian)	A Pigouvian tax of EMC per unit makes sellers pay the full social cost, so the market makes Q*. Helpful externality: a Pigouvian subsidy of EMB per unit.
Pigouvian tax on a monopoly?	It raises MC, so the monopoly cuts Q further below Q* and DWL grows.
What is total surplus?	Add CS, PS and tax revenue, then subtract the total harm to others.

Number problems with straight lines	
• Terms: Demand is $P = a - bQ$ and supply is $P = c + dQ$. Q1 is the quantity before a policy, Q2 after.	

Task	How
Equilibrium	Set demand price equal to supply price and solve for Q, then plug Q into either equation for P.
CS and PS	Each is a triangle: $CS = \frac{1}{2} \times Q \times (a - P)$ and $PS = \frac{1}{2} \times Q \times (P - c)$.
Tax t	Buyers pay t more than sellers get, so solve $a - bQ = c + dQ + t$ for Q2. Revenue = $t \times Q2$. DWL = $\frac{1}{2} \times t \times (Q1 - Q2)$.
Subsidy s	Solve $a - bQ + s = c + dQ$ for Q2. Cost = $s \times Q2$. DWL = $\frac{1}{2} \times s \times (Q2 - Q1)$.
Monopoly	Set $MR = a - 2bQ$ equal to MC to get Qm, then put Qm into demand for Pm. Given Pm, Qm and MC instead: $a = 2 \times Pm - MC$, $b = (Pm - MC) \div Qm$.
Externality	Q* solves demand = supply + EMC; Q1 solves demand = supply. DWL = $\frac{1}{2} \times EMC \times (Q1 - Q^*)$ if EMC is constant.

Check: $CS + PS + \text{revenue} + DWL$ must equal the old total.